RETAIL FLOW

PRESENTED BY :
Santiago Campos, Javier Hernández, María Parra, Ashley Harris and Carlota Prósper

2026

YOLO PREDICTS BOUNDARY BOXES

We will utilize ML YOLO to section regions in the store and pinpoint the exact location of the clients so we can separate them from other entities.

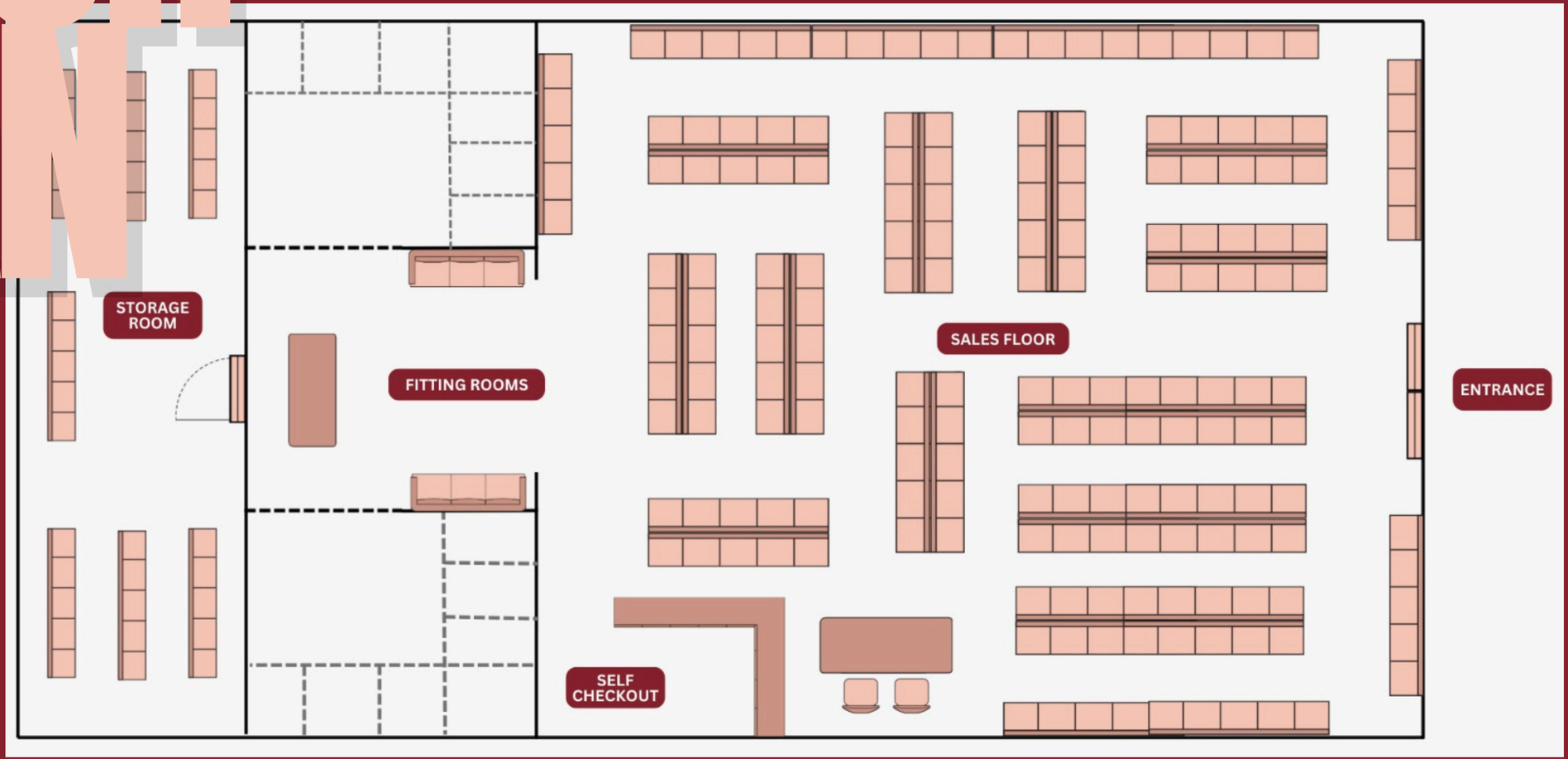
WE TRACK EACH ZONE'S STATS

We will measure the number of users detected in each zone and the duration of time allocated to each video. .

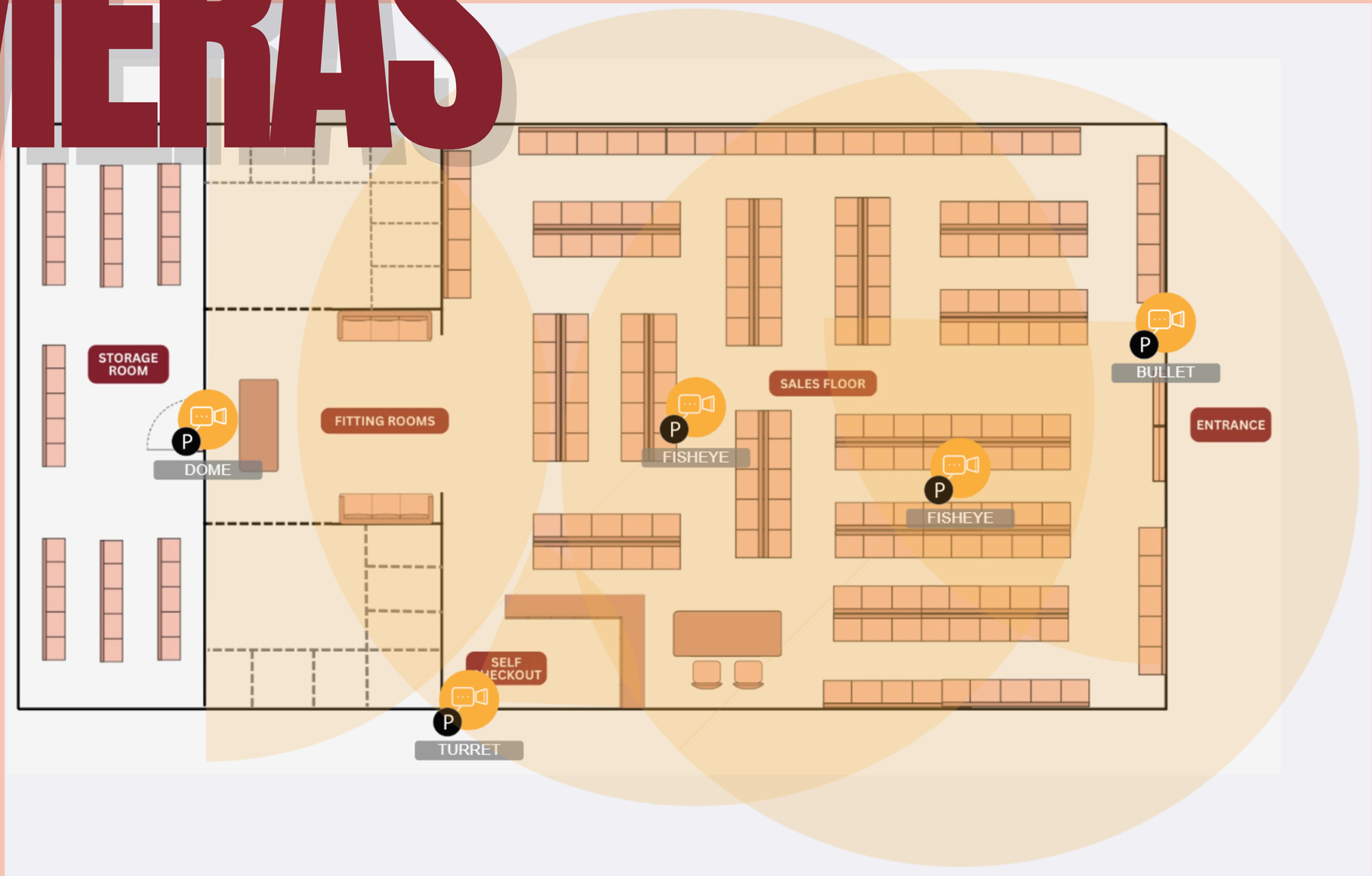
WE WILL LAUNCH AND TRACK THE MODEL

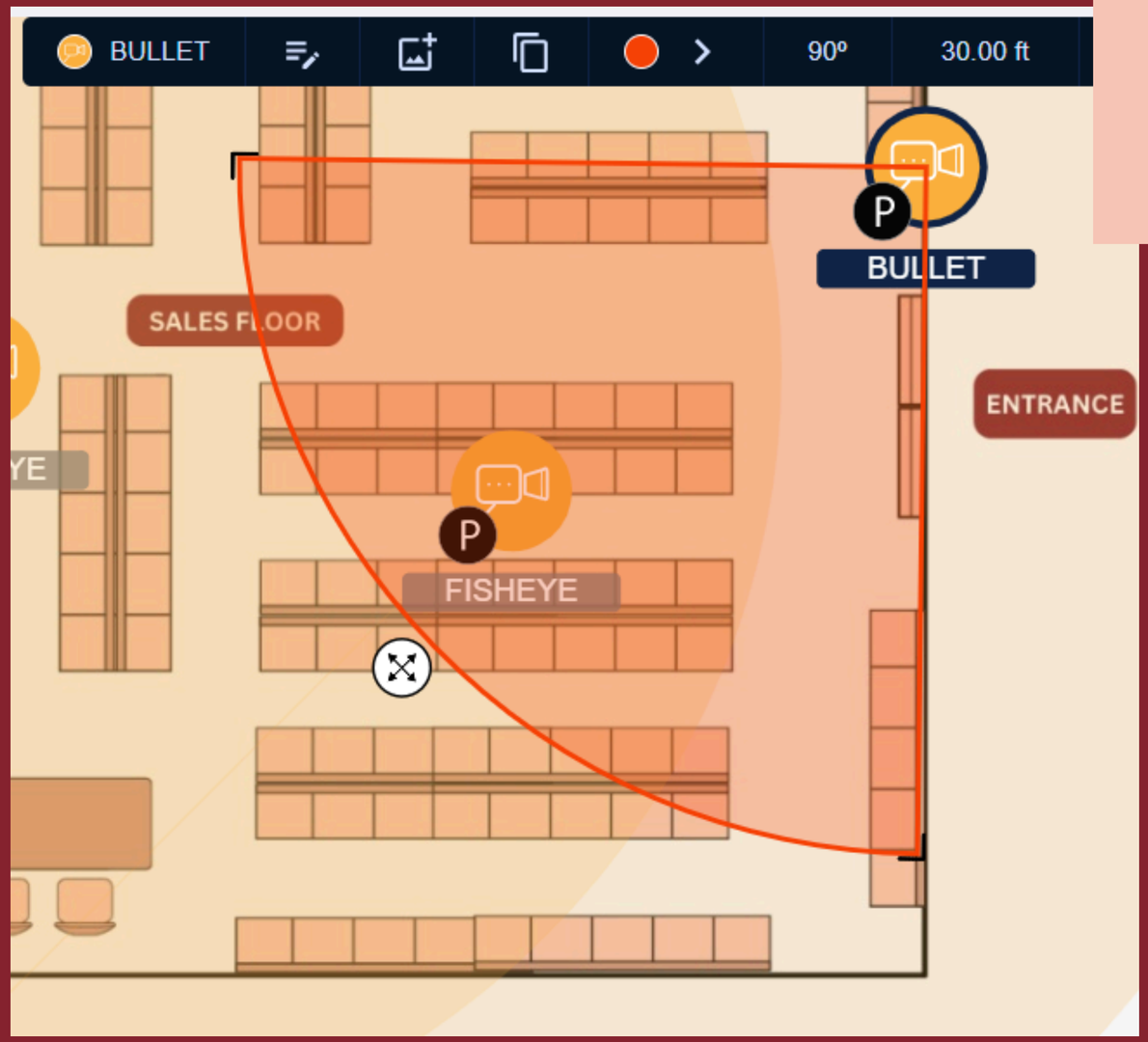
We will launch our model to the market in an adaptable format and keep track of the results.

FLOOR PLAN


STORAGE ROOM
FITTING ROOMS
SALES FLOOR
ENTRANCE
SELF CHECKOUT

CAMERAS


STORAGE ROOM
FITTING ROOMS
SALES FLOOR
ENTRANCE
SELF CHECKOUT
DOME
FISHEYE
FISHEYE
BULLET
TURRET



BULLET

VIEWING RANGE:

90º angle | 30 ft radius

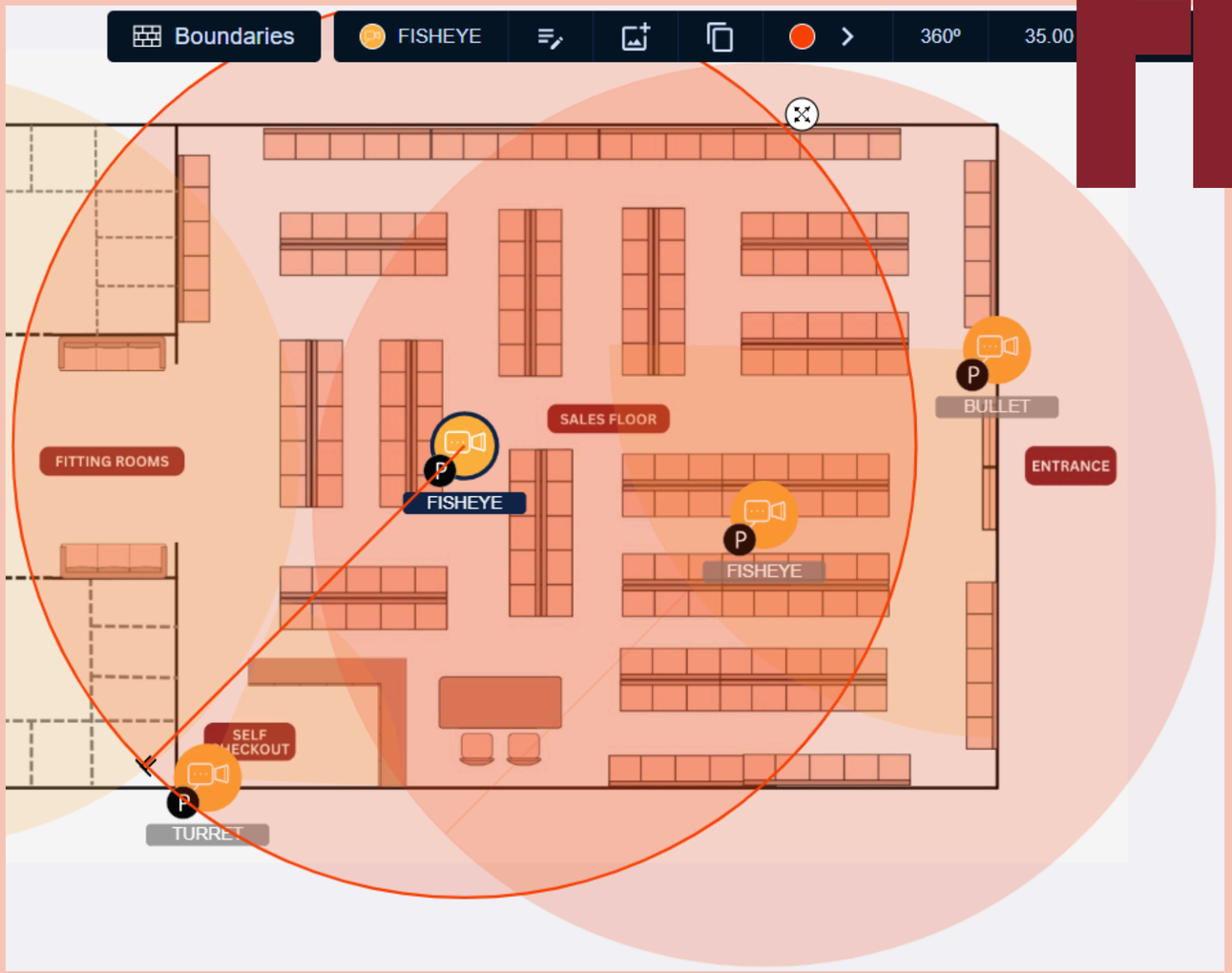
PLACEMENT:

Above primary entrance.

PURPOSE:

Record entry/exit.

FISHEYE



VIEWING RANGE:

360º angle | 35 ft radius

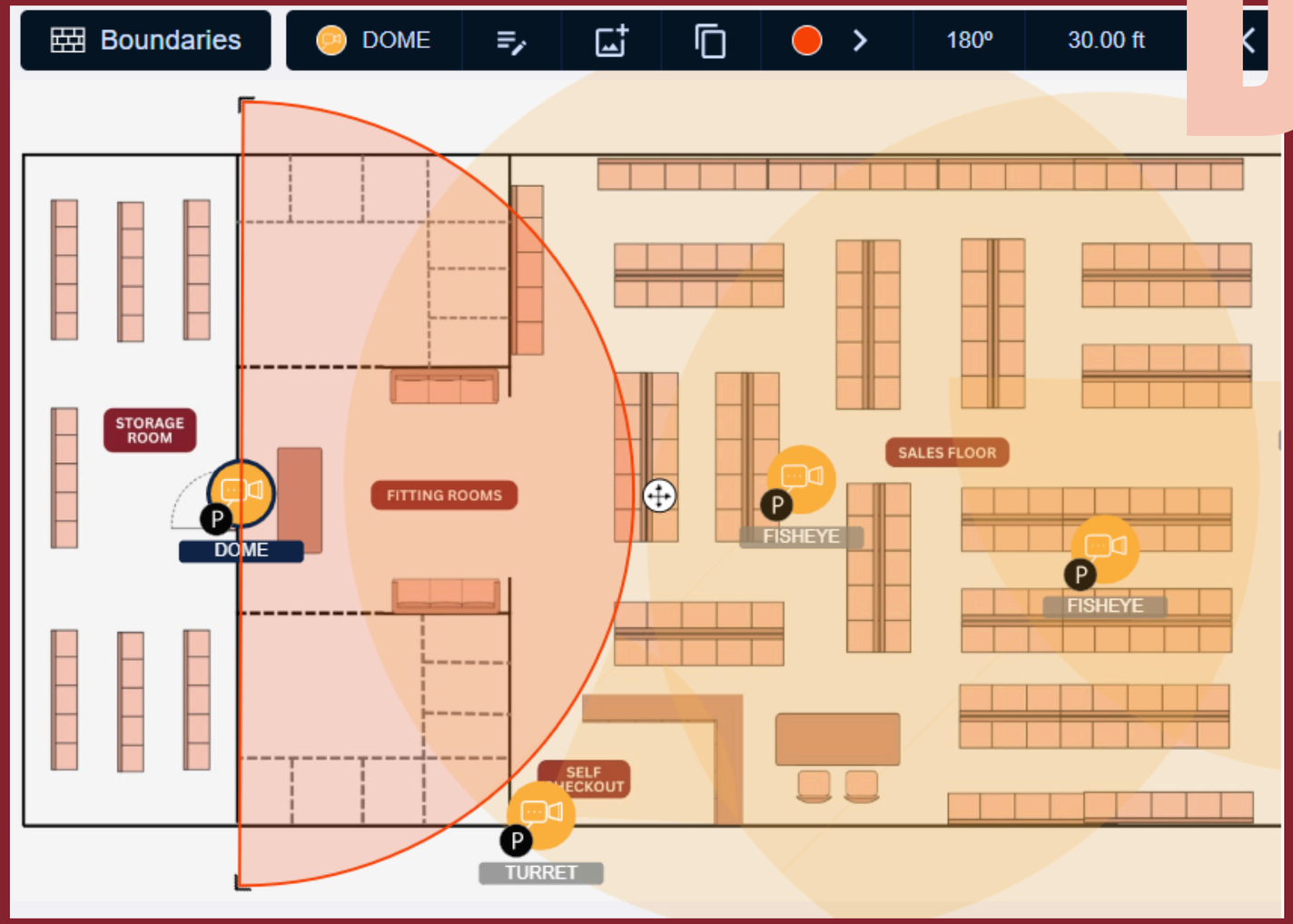
PLACEMENT:

Main sales floor.

PURPOSE:

Seamless path tracking and heat mapping.

DOME



VIEWING RANGE:

90º angle | 30 ft radius

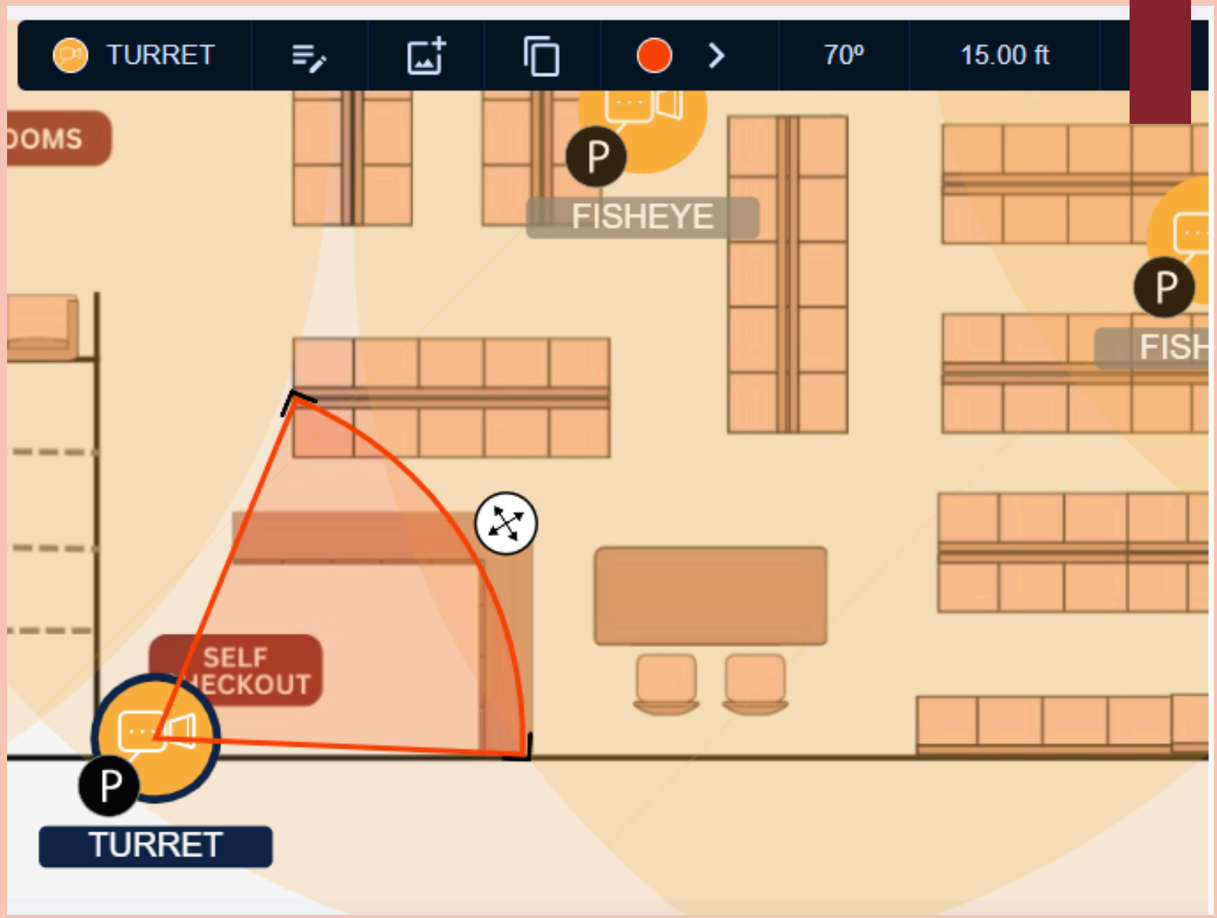
PLACEMENT:

Fitting room lobby.

PURPOSE:

Measure transitions and dwell time.

TURRET



VIEWING RANGE:

70º angle | 15 ft radius

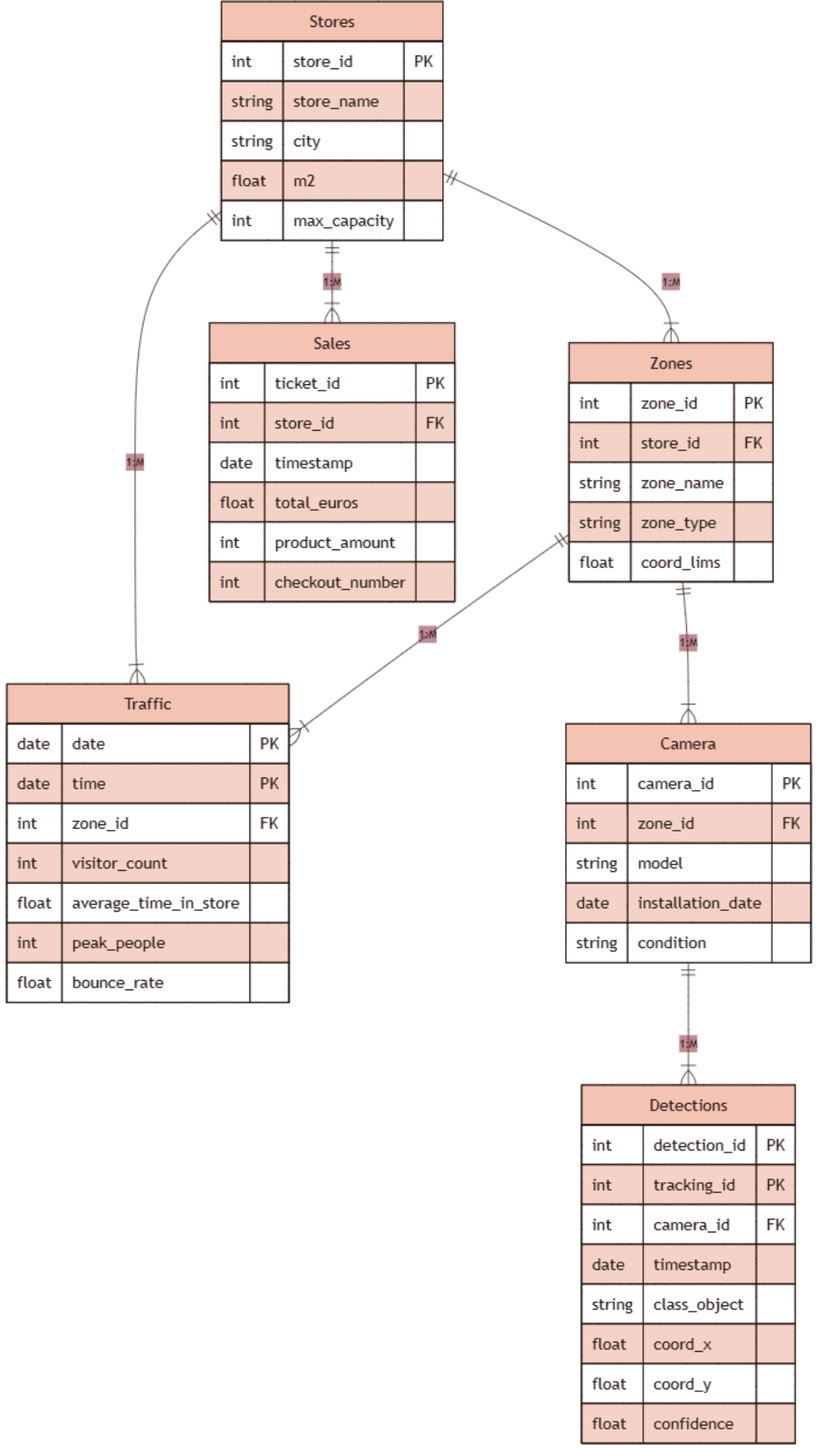
PLACEMENT:

Above self-checkout

PURPOSE:

Measure “Popularity vs Purchase”

ER DIAGRAM


Stores
int store_id PK
string store_name
string city
float m2
int max_capacity
1:M
1:M
1:M
Sales
int ticket_id PK
int store_id FK
date timestamp
float total_euros
int product_amount
int checkout_number
Zones
int zone_id PK
int store_id FK
string zone_name
string zone_type
float coord_lims
1:M
1:M
Traffic
date date PK
date time PK
int zone_id FK
int visitor_count
float average_time_in_store
int peak_people
float bounce_rate
Camera
int camera_id PK
int zone_id FK
string model
date installation_date
string condition
1:M
Detections
int detection_id PK
int tracking_id PK
int camera_id FK
date timestamp
string class_object
float coord_x
float coord_y
float confidence

VARIABLE EXPLANATION

Variables			
Stores			
Name	**Key**	**Type**	**Real world description**
store_id	PK	int	Unique identifier for each physical store location.
store_name		string	Commercial name of the branch (e.g., "Zara Gran Via").
city		string	City where the store is.
m2		float	Total surface area of the store in square meters.
max_capacity		int	Legal maximum occupancy allowed.
Sales			
ticket_id	PK	int	Unique identifier for the transaction (receipt).
store_id	FK	int	Foreign Key linking the sale to a specific Store.
timestamp		date	Exact date and time the transaction was completed.
total_euros		float	Total monetary value of the purchase (Revenue).
product_amount		int	Total quantity of individual items in the basket.
checkout_number		int	Identifier of the specific POS/Register used.
Zones			
zone_id	PK	int	Unique identifier for a logical area within the store.
store_id	FK	int	Foreign Key indicating which Store this zone belongs to.
zone_name		string	Descriptive name (e.g., "Fitting Rooms", "Entrance").
zone_type		string	Category of the area (Sales, Transit, Service, Staff).
coord_lims		float	Spatial coordinates defining the zone boundaries on the map.
Camera			
camera_id	PK	int	Unique hardware identifier for the device.
zone_id	FK	int	Foreign Key indicating which Zone the camera monitors.
model		string	Hardware model or manufacturer of the sensor.
installation_date		date	Date when the device was activated.
condition		string	Operational status (Active, Maintenance, Error).
Detections			
detection_id	PK	int	Unique ID for every single detection event.
tracking_id	PK	int	Anonymized Hash representing a unique moving object.
camera_id	FK	int	The specific camera that generated this detection.
timestamp		date	High-precision timestamp (ms) of the detection.
class_object		string	Classification of the object (Person, Staff, Cart).
coord_x		float	Normalized X-coordinate position on the floor plan.
coord_y		float	Normalized Y-coordinate position on the floor plan.
confidence		float	AI Probability score (0.0 - 1.0) of the classification.
Traffic			
date	PK	date	Date of the record (YYYY-MM-DD).
time	PK	date	Hour slot of the record (0-23).
zone_id	FK	int	The specific Zone being analyzed.
visitor_count		int	Total number of unique tracking_ids detected.
average_time_in_store		float	Average dwell time (seconds) per visitor.
peak_people		int	Maximum number of simultaneous people detected.
bounce_rate		float	Percentage of visitors who leave after entering the store without interacting.

DATA SOURCE

- Sales: originates from the physical cash registers.
- Detections & traffic: originates from the cameras.
- Static configuration: defined by the business layout.

THANK YOU

2026